SCREEN TIME AND BMI PERCENTILE

SLR

Simple Linear Regression using YRBS 2007 | $n = 12{,}844$ | $\alpha = 0.05$

1 RESEARCH QUESTION

Is screen time associated with BMI percentile among students?

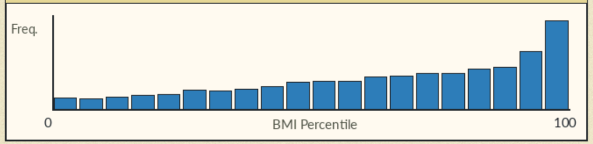
2 DATA & VARIABLES

- **Dataset:** YRBS 2007
- **Sample size:** n = 12,844
- **X variable:** Screen time hours
- **Y variable:** BMI percentile

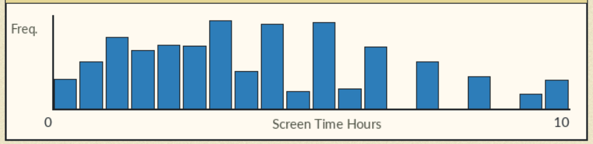
3 EDA HIGHLIGHTS

- BMI percentile spans nearly the full 0-100 range, with many observations near the higher percentiles.
- Screen time is concentrated at low to moderate hour values, with fewer students at very high hours.
- The raw scatter suggests only a weak upward trend.

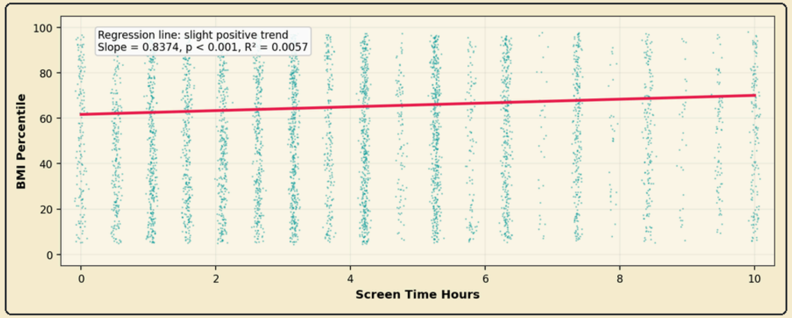
BMI PERCENTILE DISTRIBUTION



SCREEN TIME HOURS DISTRIBUTION



4 MAIN ANALYSIS



Each point represents one student. The fitted line shows a slight positive association.

5 MAIN RESULTS

Parameter / Metric	Estimate	t Stat	p-value	95% CI	R^2
Intercept (β_0)	61.6837	–	–	–	–
Slope, screen hours (β_1)	0.8374	8.6015	< 0.001	[0.6466, 1.0282]	0.0057

ŷ **Fitted line equation:**
Predicted BMI percentile = 61.68 + 0.84 × screen time hours

6 METHOD

β **Simple linear regression**

Model: BMI percentile = $\beta_0 + \beta_1$(screen time hours) + ε

Goal: test whether β_1 is different from 0.

7 KEY NUMBERS

SLOPE	P-VALUE	R^2
0.8374 BMI points / hour	**<0.001** statistically significant	**0.0057** only 0.57% explained

8 INTERPRETATION / TAKEAWAY

- Each additional hour of screen time is associated with about 0.84 points higher BMI percentile on average.
- The association is statistically significant, but the effect is weak in practical terms.
- R-squared = 0.0057 means screen time explains only 0.57% of the variation in BMI percentile.

9 DIAGNOSTICS / LIMITATIONS

- Residual diagnostics suggest deviations from perfect normality.
- BMI percentile and screen time are both discrete or limited-format measures.
- Screen time is self-reported.
- Observational data show association, not causation.

10 CONCLUSION

Σ

- There is a positive association between screen time and BMI percentile.
- The slope is statistically significant, but the explanatory power is very low.
- Screen time alone is not a strong predictor of BMI percentile.

Takeaway: The results are statistically significant but practically weak positive association.

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